

**Economic Implications of Rising Military
Burdens:
An Outlook for Asia-Pacific Countries**

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Abstract

The Asia-Pacific region benefited from three decades of relative peace following the end of the Cold War, enabling exceptional economic growth through the reallocation of fiscal resources away from defense toward productive investment, the so-called peace dividend. Since 2014, however, the intensification of geopolitical tensions has driven a sustained increase in military burdens across the region, raising the risk that these gains will be eroded at precisely the moment when member states face new development and demographic challenges. This paper examines the economic implications of this trend. First, it reviews the academic literature on the military spending-growth nexus, which yields no uniform verdict but shows a growing weight of evidence toward negative effects, particularly in lower-income and non-arms-exporting economies. Second, it documents the trajectory of military burdens and conflict intensity across Asia-Pacific member states from 1980 to 2024, highlighting the emergence and subsequent erosion of the peace dividend. Third, it presents panel econometric evidence using a Two-Way Fixed Effects (TWFE) specification on Asia-Pacific countries over 1993-2024. The aggregate estimated effect is negative but statistically non-significant ($\hat{\beta} = -0.122$), reflecting deep regional heterogeneity. Disaggregated results reveal a statistically significant negative effect of -0.293 percentage points for non-arms-exporting countries, the majority of Asia-Pacific states, while major arms exporters experience a net positive effect ($+0.637$ percentage points) through industrial spillovers and export revenues. For conflict-affected countries, military spending carries a positive and significant effect during active conflict years, operating through demand stabilisation and security restoration channels. These findings support differentiated policy recommendations: minimising the fiscal cost of defense for net importers, investing in civilian innovation ecosystems for industrial exporters, treating conflict prevention as a first-order economic priority, and promoting cross-border defense-industrial interdependence to reduce incentives for unilateral military escalation.

Keywords: military spending; military burden; peace dividend; economic growth; Asia-Pacific; panel data; two-way fixed effects; arms exports; geopolitical risk

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1. Introduction

The relationship between security and economic development is one of the most important policy questions that Asia-Pacific governments are facing today. Since the end of the Cold War, the region experienced a prolonged period of relative peace. The decline of the military burden, meaning military expenditures as a percentage of GDP, freed fiscal resources for the investments that allowed Asia's exceptional growth.

However, since 2014, the intensification of territorial disputes in the South China Sea, renewed tensions on the Korean Peninsula, Russia's invasion of Ukraine, US–China rivalries, Myanmar's civil war, rising tensions between India and Pakistan, Pakistan and Afghanistan, and the ongoing conflict in Iran have disturbed the region's peace. Military burdens have consequently experienced a ripple effect and spiraled upward, forming part of the aggregate continuous increases in military spending since 1989 documented by the United Nations Secretary-General (2024). The economic gains of decades of relative peace, the so-called peace dividends, are at risk of being eroded precisely at the moment when the region faces new development and demographic challenges (ESCAP, 2026).

For more than thirty years, peace dividends materialised in the Asia-Pacific region, allowing for its significant development. The opportunity cost of war, trade disruptions and sanctions, reduced investment, and the contraction of fiscal space have historically served as a powerful deterrent to conflict in the region. Indeed, the prospect of economic growth through openness drove East Asia toward peace after 1979.¹ China's post-cultural revolution reforms, Viet Nam's Doi Moi, and the Soviet Union's perestroika all reflect a turn toward development rather than confrontation (Tønnesson, 2015). This peace enabled a reallocation of public resources away from defense toward productive investment in infrastructure, human capital, and industrialisation. Japan's experience is emblematic: by capping defense spending at 1% of GDP and benefiting from the United States' security umbrella, it channelled newly available fiscal resources into the industrial and technological investments that sustained its postwar growth miracle (Hook and Spanier, 2011; SIPRI, 2023). Focusing on the peace dividends theory, Mayberry (2023) produced a global analysis and found positive results. This economic logic (peace allowing for greater growth) is also supported by an IMF (2024) cost-benefit analysis: every dollar invested in conflict prevention can save up to 103 dollars in conflict-related costs.

Defense expenditures affect development through multiple, and sometimes opposing, transmission channels, making their net impact a central empirical question. On

¹The last interstate war between China and Viet Nam occurred in 1979.

one hand, military spending can crowd out productive investment in infrastructure and human capital, accumulate public debt in fiscally constrained economies, or deter foreign direct investment by signalling instability. On the other hand, it can stimulate domestic demand, generate technological spillovers from military to civilian industry, or lead to export revenues for producing countries. The United Nations Secretary-General (2024) underscores the urgency of this question: global military expenditure reached USD 2.7 trillion in 2024, while the annual financing gap for the Sustainable Development Goals (SDGs) stands at USD 4 trillion. This illustrates the peace and stability needed to concentrate on economic development and achieve the UN's SDGs (Dutt, 2024).

This paper examines the dynamics between military burden and economic growth in the Asia-Pacific region through three complementary lenses. Section 2 reviews the academic literature on the relationship between military spending and economic growth. Section 3 documents the Asia-Pacific region's relative peace since 1990, the benefits of the subsequent peace dividends, and their erosion since 2014. Section 4 presents the data and empirical methodology, while Section 5 explores the heterogeneity of the relationship between military burden and economic growth through an econometric analysis of Asia-Pacific member states. Section 6 discusses the main findings, Section ?? draws policy recommendations, and Section 7 concludes.

2. Literature Review: Military Spending and Economic Growth

The literature on military spending and economic growth delivers no clear verdict: the sign, magnitude, and direction of the effect all depend on transmission channels, spending composition, and national context. Several meta-analyses compiled decades of empirical work and showed that positive, negative, and null results coexist across studies. However, a recent shift toward a more negative effect has emerged in recent years, especially when looking at the post-Cold War period. In this literature, military spending affects growth negatively through four main channels: crowding-out of productive investment, displacement of social expenditure, inflationary pressure, and FDI deterrence. Some studies, concentrated on high-income arms-exporting economies and conflict-affected countries, identify positive effects through demand stimulus and defense-to-civilian knowledge spillovers. Yet a growing majority find net negative effects, mainly driven by the crowding-out of productive investment, hidden unemployment, and higher debt levels. Therefore, the sign and magnitude of the effect appear to be conditional on spending composition, financing mechanisms, national industry structure, and alliance membership. Together, these findings

suggest that drawing uniform policy conclusions can be misleading: what holds for an arms-exporting country tells us little about a developing net importer facing fiscal constraints.

2.1 Higher Military Expenditure and Economic Growth: The Case for Positive Effects

The earliest argument for a positive relationship between military spending and economic growth emerged during the Cold War (Benoit, 1978). This finding generated a large empirical literature that explores three main theoretical channels through which military spending might stimulate growth.

First, an increase in military spending leads to higher aggregate demand which, under conditions of spare capacity, generates a multiplier effect, on average close to one (IMF, 2026). For example, an increase in military personnel would lead to higher demand for military equipment, weapons, vehicles and bases. As domestic businesses move to meet this demand, unemployment falls, profits rise, and government tax revenues increase, more than offsetting the initial public spending and resulting in a higher overall GDP. This Keynesian demand channel has been observed in India, which launched the 2014 “Make in India” initiative, aiming to manufacture defense equipment domestically to take advantage of this channel. However, its efficiency is greatly heterogeneous among different economies, being especially conditional on whether or not the government uses the domestic industry to fulfil public contracts.

The literature has therefore documented positive effects in some developed countries until 1990 (Alptekin and Levine, 2012). These countries often benefit from an important industrial-military complex and capital-intensive armies, countries whose military budgets are spent primarily on equipment and technology rather than salaries, allowing them to transform these expenditures into R&D spillovers. The R&D spillover channel has received the most rigorous recent empirical support: a 10% increase in public military R&D generates a 4.3% increase in private R&D, with confirmed international spillovers depending on military alliance structure (Dimitriou et al., 2024; Moretti et al., 2025).

This crowding-in effect, defense spending attracting rather than displacing productive investment, requires a developed civilian innovation ecosystem capable of absorbing and commercialising defense-generated knowledge. A strong private sector was capable of turning defense technologies such as GPS and the internet, both originally developed for military purposes, into commercial products. Moreover, such industry can benefit from economies of scale: producing 1,000 fighter jets is cheaper per unit than producing 10.

Another positive channel operates through arms exports. When geopolitical rivalry increases, arms-exporting countries tend to benefit from positive externalities of increasing their military expenditures (Callado-Muñoz et al., 2022; Dimitriou et al., 2025; Rahman and Siddiqui, 2019). Japan recently took this path by allowing for greater arms exports, including US Patriot missile components, to non-conflict-affected territories. However, these achievements can only be met by very few Asia-Pacific member states.

Finally, the security channel remains relevant. Higher military spending signals instability to foreign investors but can also attract investment by demonstrating a credible defense posture in the context of regional threats (Dunne and Smith, 2020; Nugroho and Purwanti, 2021). For the Republic of Korea, high military spending coexisted with strong economic growth thanks to a favourable development environment: high returns on productive investment, rapid human capital accumulation, and strong catch-up potential through technology transfer. Samsung and Hyundai, for example, developed capabilities through defense contracts. These factors can help to offset the negative effects that military expenditure has on growth, though high spending can also signal a threat that deters investment.

2.2 Higher Military Expenditure and Economic Growth: The Case for Negative Effects

The weight of more recent evidence has shifted decisively toward a negative relationship between military expenditure and economic growth. A meta-analysis of studies published between 1973 and 2013 finds that only 23% report a positive effect. The share of studies reporting a negative effect rose to 55% among those published after 2006 (Dunne and Tian, 2013). The meta-analysis of Simpartl (2024) estimates a fixed-effects mean of -0.147 , meaning that on average, a one-unit increase in military burden is associated with a 0.147 percentage point reduction in GDP growth. This reflects a post-Cold War crowding-out effect: after the highly militarised Cold War period, at least until 1980, military spending reduces the ability of a government to allocate funds to growth-enhancing sectors.

Beyond human capital, higher military spending can crowd out private investment through a financing channel (Malizard, 2015). When governments fund defense increases by issuing bonds, they compete directly with the private sector for available savings. As public debt instruments offer perceived security and attractive returns, private investors may shift their funds away from productive business investment, leaving firms with less access to capital precisely when economic dynamism is most needed.

The negative growth impact is also driven by personnel spending rather than capital expenditure (Dimitriou et al., 2025), as personnel-intensive armies can hide unemployment issues (Becker, 2021; Becker and Dunne, 2021). They can mask underlying unemployment problems by absorbing surplus labour into the military without generating productive economic returns. This distinction is particularly relevant for lower-income Asia-Pacific states, where rearmament is likely to be personnel-intensive given fiscal and industrial constraints. Alliance membership also matters: NATO partially buffers the negative effect through burden-sharing, whereas SCO and CSTO provide no equivalent offset (Dimitriou et al., 2024). Most Asia-Pacific countries belong to neither NATO nor any alliance with comparable burden-sharing mechanisms, leaving them more exposed to the full economic cost of rearmament.

Pakistan illustrates the difficult trade-offs that military spending can impose on lower-income economies. In the 2025-26 federal budget, defense received 14.5 per cent of total government expenditure, while higher education and health were allocated only 0.2 and 0.1 per cent respectively (ISSI, 2025). With debt servicing consuming nearly half of total expenditure, every additional rupee allocated to defense directly competes with investment in human capital and social infrastructure. Unlike the Republic of Korea, Pakistan has not been able to translate its military expenditure into significant industrial or technological spillovers, and remains the top Asia-Pacific arms importer in 2022-24 (0.41% of GDP), leaving the economy exposed to the crowding-out effect without capturing the offsetting benefits.

Awaworyi Churchill and Yew (2018) showed that the negative effect is concentrated in lower-income countries, where military budgets tend to be financed through monetary expansion or debt in already fiscally constrained economies. These countries also often lack a strong military industry which would allow them to benefit from R&D spillovers and arms exports. The most credibly identified estimate in the literature is provided by Saeed (2025), who used arms imports and the number of neighbouring states experiencing interstate violence as instruments.² As the regional environment is one of the main mechanisms at play within the Asia-Pacific tandem increase, the estimated effect of -1.1 percentage points³ provides a useful benchmark for policy recommendations (United Nations Secretary-General, 2024).

²Using indirect statistical techniques to isolate causality.

³Meaning that an increase in military expenditure as a share of GDP (military burden) of 1 percentage point reduces economic growth by 1.10 percentage points.

2.3 Historical Precedent: Lessons from the Cold War Arms Race

The Cold War arms race offers a historically instructive precedent for the relationship between military burden and long-run economic performance. The 1981 Reagan military expansion is frequently identified as a decisive factor in the Soviet Union's economic exhaustion and collapse, as it may have led the USSR to raise its military burdens to an unsustainable level in response. However, extensive analysis by Chernoff (1991) showed an already existing high military burden before 1981. Thus, the USSR collapse, while being fuelled by lack of fiscal resources, was not a consequence of US military policy.

Quantifying the Soviet military burden is methodologically challenging. Very few reliable estimates are available. Official numbers were falsified to protect Soviet interests: in May 1989, during the Glasnost era, Gorbachev revealed that the real 1989 Soviet military expenditure level was three times higher than the falsified official figure announced in 1988. Later, in his memoirs (Gorbachev, 1996), he disclosed that before his tenure, only three to four members of the Politburo were aware of the real defense figures. Beyond these deliberate concealments, the planned economy is structurally different from a free market one, and the purchasing power in roubles is difficult to compare to the dollar equivalent, as official price levels were manipulated. The Soviet Union benefited from inexpensive access to industrial goods and labor, allowing it to convert a smaller budget (in constant USD) into more military equipment.

In 1975, a declassified CIA report, even if later criticized by Lee (1995), provided a reliable lower bound for the Soviet military burden. While the United States and Soviet burdens were broadly similar (around 6% of Gross National Product), the Soviet defense sector accounted for 22% of machinery production, against only 10% in the US. The working-age population employed in the military and defense sector reached 6% in the USSR against 3% in the US, revealing a highly militarised society. More recent estimates (Harrison, 2003; SIPRI, 1998) place the burden at between 7.9% of GNP in 1980 and 8.4% of GNP in 1987. The most credible estimate comes from Gorbachev's own memoirs, which indicates that military expenditure represented up to 40% of the state budget and 20% of GNP in the 1980s, a military burden roughly twice that of the United States, while absolute military expenditures were broadly comparable after 1975 (Ricón, 2016).

Even based on the lower bound, the military burden deprived socialist republics of needed investment while crowding out the workforce. One CIA analysis (CIA, 1975) points out that even if the defense sector helped the Soviet industry to develop electronics and nuclear power, the crowding-out of high-grade scientific capacity broadly reduced innovation. Limited export capacity and market closedness also

prevented Soviet innovations from being reproduced and commercialised in a growth-enhancing way. Thus, the military burden was unsustainably high while R&D spillovers were structurally too low to offset the negative effects.

The United States maintained a military burden between 10% and 14% of GDP during the 1950s, particularly during the Korean War, before declining to approximately 5% during the 1970s. The 1981 Reagan military buildup reversed this trend, raising the military burden to 6.81% of GDP in 1982 (Liang et al., 2025). However, the net impact on growth was mixed. Hou et al. (2025) analysed the 1972-2021 period and found that US military spending broadly constrained economic growth, except when looking at military R&D expenditure: defense-funded research generated civilian spillovers through technologies such as the internet, the GPS, and semiconductors. This channel is empirically confirmed by Moretti et al. (2025), who estimate that over 1987-2019, a 10% increase in public military R&D generated a 4.3% increase in private R&D.

The contrast with the Soviet Union is particularly instructive. Both superpowers spent comparable amounts in absolute terms after 1975. However, the Soviet burden represented nearly twice the share of national income. The military burden effect on growth appears to be non-linear and negative after a certain threshold (Dunne and Makanza, 2019); thus, an increase in military spending can harm economic growth more severely in the USSR than in the US. Moreover, the planned economy was less efficient in capturing R&D spillovers through commercialisation and exportation. The economic cost of military spending therefore appears to depend on the institutional capacity to convert defense investment into productive civilian activity, a lesson directly relevant to certain Asia-Pacific economies. One should keep in mind that the Soviet military burden also fuelled the heavy industry, provided work to a large part of the population, and promoted the allocation of highly-qualified workers. Nonetheless, no one can dispute the crowding-out effect of such a high military burden, especially on the investments the USSR needed to modernise its industry and innovation capacity in the late 1970s.

3. The Asia-Pacific Region: From Peace Dividends to Rising Tensions

3.1 Three Decades of Relative Peace

Asia and the Pacific enjoyed three decades of relative peace following the end of the Cold War, enabling remarkable economic success, but this favourable environment is now under threat. Since the 1990s, Asia-Pacific member states have experienced

a period of relative peace compared to the rest of the world. When measured by battle-related deaths per million inhabitants, Asia-Pacific countries systematically recorded lower values than other world regions on average. While Sub-Saharan Africa and the Middle East suffered large-scale wars, most of Asia and the Pacific remained largely insulated from large-scale interstate violence. This does not mean that armed conflicts were absent, but rather that their human cost per capita was significantly lower than elsewhere.

This relative advantage is not uniform across the region. South Asia in particular is the theatre of repeated confrontations. India and Pakistan, both nuclear-armed states, sustained a chronic state of tension along their shared border, especially in the Kashmir region. Afghanistan endured decades of continuous conflict, from civil wars and international military presence until 2021, to an open conflict with Pakistan in 2026. As a result, the share of Asia-Pacific countries formally “at war”, using the standard threshold of 1,000 deaths a year (UCDP), has been the highest worldwide between 1990 and 2010. Yet the intensity of violence in per capita terms remained lower than in other conflict-affected regions. Indeed, the share of Asia-Pacific countries at war peaked at around 15 per cent in 1992, following the collapse of the USSR, and fell to as low as 3 per cent in 2018,⁴ while keeping battle-related deaths per capita relatively constant. This broad trajectory of decreasing conflict exposure proved to be one of the drivers of the region’s economic success.

Figure 1. Battle Deaths Per Million by World Region, 1990-2024
3-year centred rolling average | UCDP/World Bank
ESCAP members = thick solid black | Other regions exclude ESCAP countries

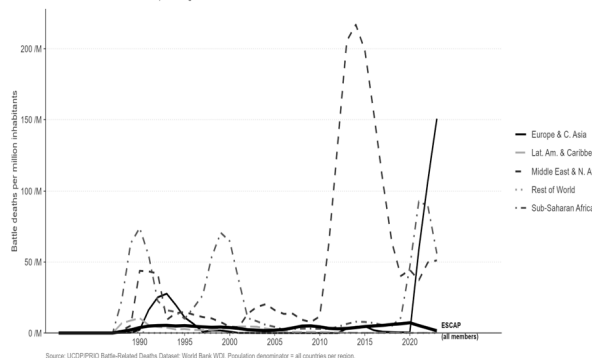
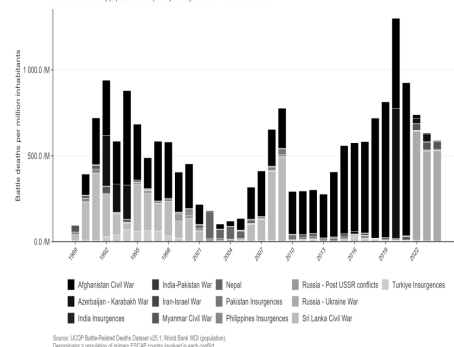


Figure 2. Battle Deaths per Million Inhabitants by Conflict, ESCAP Member States, 1989-2024
Deaths normalised by population of the primary country affected – UCDP best estimate



3.2 The Benefits of the Peace Dividend

The reduction in military burden that followed the end of the Cold War freed up fiscal resources that directly contributed to the region’s growth miracle. The end of the Cold War triggered what economists call a “peace dividend”, the economic gains that arise when governments reduce their military burden⁵ and redirect those resources toward more productive uses. In practical terms, this means that money previously spent on weapons, soldiers, and military infrastructure can be reallocated

⁴ Author’s calculations from UCDP data.

⁵ Known as military expenditure as a share of GDP.

to health, education, and public infrastructure, as well as to private investment (IMF, 1996; Mayberry, 2023). Across Asia and the Pacific, this shift was substantial. Cambodia and Viet Nam, having emerged from devastating conflicts in the preceding decades, became two of the most dynamic and fast-growing economies in the region (Hill and Menon, 2014).

From 1980 to 2014, the GDP-weighted military burden declined across Asia-Pacific countries. The resources freed by this reallocation were redirected toward development-friendly investments. This period of declining military burdens broadly coincided with what is referred to as the region’s “growth miracle”, decades of exceptional economic expansion and development that greatly reduced poverty (World Bank, 1993, 2013).

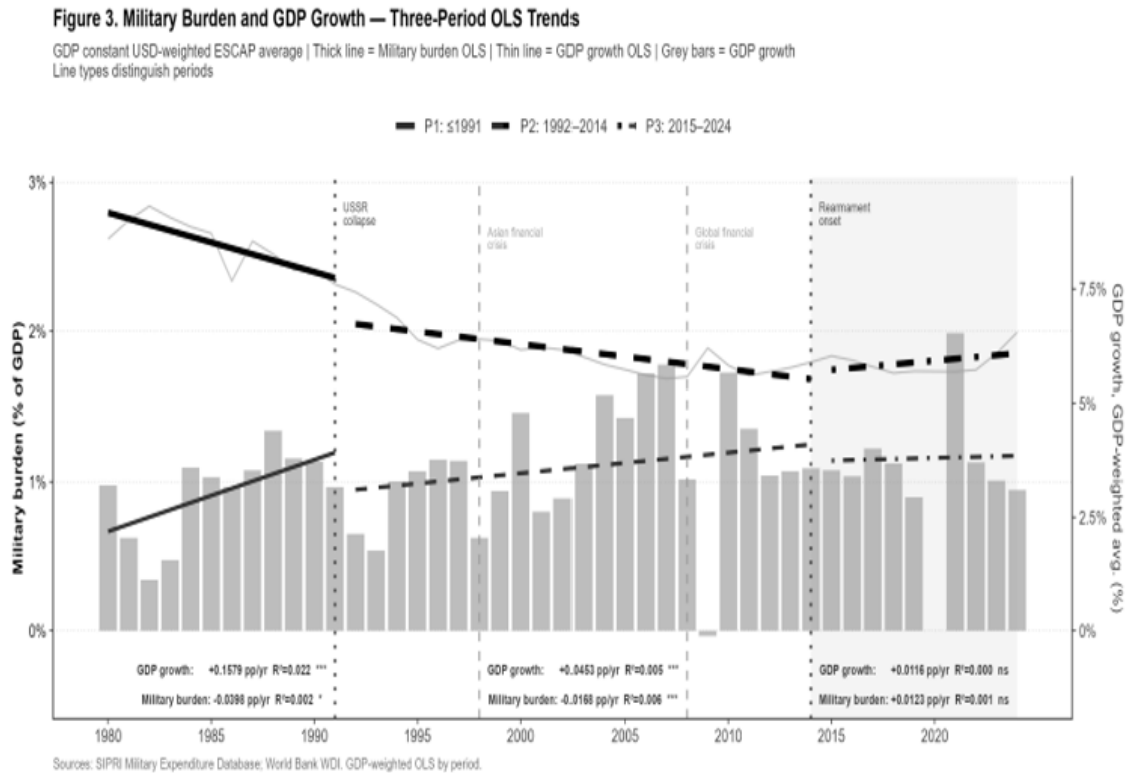
The correlation between declining military burdens and rising growth rates across Asia-Pacific countries is visible in the data. Periods of falling military burden track closely with periods of GDP-weighted constant-USD growth. A clear pattern emerges for both the 1980-1991 and 1992-2014 periods: as military burdens fell, GDP growth accelerated. Even if this correlation is not sufficient to establish causality, as multiple structural factors and global financial crises⁶ affect growth, it remains consistent with the theoretical prediction that higher military spending crowds out resources from higher-return investments (Dunne and Tian, 2013; Simpartl, 2024). However, after 2008, the military burden and economic growth trends both became statistically indeterminate: when the military burden is no longer decreasing, economic growth is also no longer consistently increasing. This effect has various explanations and is heterogeneous across countries.

The historical cases of Japan and the Republic of Korea are instructive in this regard. Japan illustrates how low military burdens can underpin exceptional economic performance. Following the Second World War, Japan adopted a strictly defensive security posture, and in 1976 formalised a cap at 1% of GNP (later shifted to GDP). Benefiting from the US security umbrella, Japan directed freed fiscal resources toward industrial investment, technological upgrading, and human capital formation. A low military burden was thus a permissive condition for the growth miracle (Hook and Spanier, 2011; SIPRI, 2023). This settlement is now being reversed. Driven by concerns over rising tensions in the South China Sea, North Korea’s missile programme, and Russia’s invasions of Ukraine (2014 and 2022), the 1% norm was progressively abandoned from 2017 onwards. In December 2022, Prime Minister Kishida committed to raising national-security-related spending to 2% of GDP by 2027. With public debt exceeding 250% of GDP and accelerating demographic ageing, higher defense spending will inevitably compete with social protection and productive

⁶Including the 1998 Asian Financial Crisis and the 2008 Global Financial Crisis.

investment. This position has been supported by Prime Minister Miki, who relaxed the 1976 Japanese ban on defense equipment exports in order to strengthen the country's civilian industries and technological capabilities (Wallace, 2026).

The Republic of Korea, facing significant geopolitical tensions, maintained a high military burden and yet achieved strong economic growth, thanks to its high potential for catch-up from technology transfer, which offset the depressing effect of military expenditure (Dunne and Smith, 2020).⁷



3.3 The Return of Military Spending since 2008

Driven by a combination of the 2008 global financial crisis, which altered the geopolitical calculus of major powers, and the intensification of territorial and maritime disputes across the region, the peace dividend began to erode. Military expenditure as a share of GDP stabilised across most Asia-Pacific countries after 2008 and resumed an upward trajectory after 2014. This militarisation of the region has led to years of consecutive increases in military expenditure since 2014, reaching its highest level since the Cold War (Liang et al., 2025). This rise can be explained by Russia's annexation of Crimea, the intensification of South China Sea disputes, and renewed tensions in the Korean Peninsula.

Within the Asia-Pacific, this resurgence is not occurring in isolation. The United Nations Secretary-General (2024) documents an action-reaction dynamic in East and

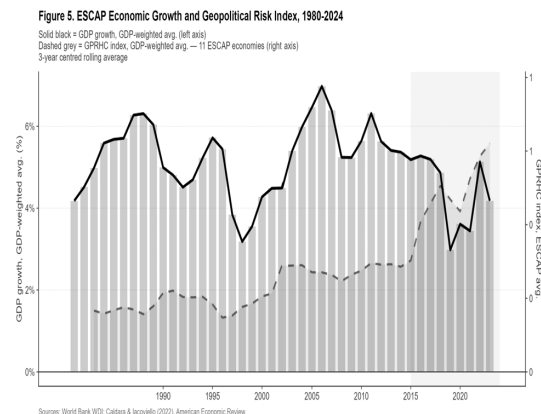
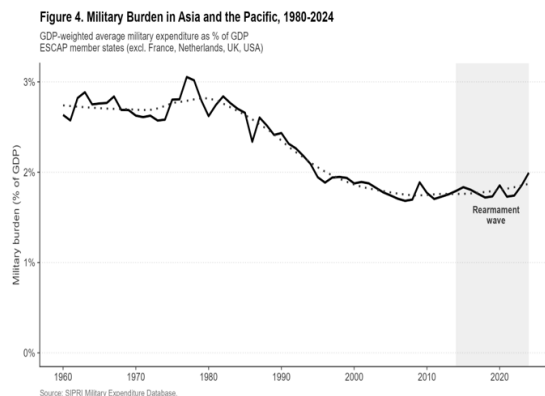
⁷Year 2020 is excluded from the analysis.

South-East Asia observable since 1989: one country's increase in military expenditure leads to similar responses from its neighbours. This dynamic of consecutive increases generates an upward spiral in regional spending which is globally welfare-reducing (Skogstad, 2016). Due to ongoing conflicts and geopolitical tensions in Myanmar (currently experiencing a civil war), between Pakistan and India, and in the South China Sea, the Asia-Pacific region is becoming one of the global hotspots, notably alongside Ukraine and the Middle East (ACLED, 2026).

The Philippines offers a telling illustration of this dynamic: facing mounting pressure in the South China Sea, Manila launched its Re-Horizon 3 military programme in 2024, committing up to USD 35 billion over the next decade. Already facing an annual deficit of over 5 per cent of GDP, this expenditure could reduce the already limited fiscal space available.

The geopolitical risk index of Caldara and Iacoviello (2022) allows us to represent these rising geopolitical tensions within Asia-Pacific countries. This index uses the occurrences of specific words such as “war” or “threat” within North American and UK newspapers to proxy the rise in geopolitical risk for a given country or world region. Computed from the available database,⁸ the GPR index closely tracks a relative deceleration in regional GDP growth in the post-2014 period, suggesting that rising geopolitical tensions are already leaving an economic footprint.

This shift carries significant opportunity costs. The annual financing gap for the Sustainable Development Goals stands at USD 4 trillion globally, while military expenditure alone has already reached USD 2.718 trillion. For Asia-Pacific governments, already facing fiscal pressures from demographic ageing and the urgent need to accelerate structural transformation, every dollar reallocated toward defense is a dollar not invested in sustainable long-term growth.



⁸Using available data for Japan, New Zealand, Singapore, Türkiye, Republic of Korea, Russia, Pakistan, India, China and Australia.

4. Data and Empirical Methodology

4.1 Sample Construction and Variables

The regression analysis covers Asia-Pacific member states over the period 1993-2024, while the descriptive analysis extends back to 1980. The starting date of 1993 reflects the fact that several Asia-Pacific countries did not officially exist prior to the collapse of the USSR, which would result in an unbalanced panel during the early transition period. Moreover, the Cold War effects of the military burden on economic growth appear different from contemporary ones. The year 2020 is excluded from all GDP-based regressions, given the mechanical distortion introduced by the COVID-19 shock on both growth rates and the military burden ratio.

Among Asia-Pacific member states, several countries could not be included in the analysis due to data limitations. Bhutan, Maldives, Samoa, Solomon Islands, Tonga, Vanuatu, and the Democratic People’s Republic of Korea are entirely absent from the regression sample, as no usable observations on military expenditure are available over the sample period. Four additional countries present in the descriptive analysis are effectively marginal in the regressions due to the high proportion of missing observations on the military burden: Turkmenistan (81 per cent of country-years missing), Uzbekistan (68 per cent), Afghanistan (43 per cent) and Viet Nam (43 per cent).

The dependent variable is the annual GDP growth rate in constant 2015 USD. The key independent variable is the military burden, military expenditure as a share of GDP, lagged by one period. Control variables follow the standard approach in the panel growth literature, as applied by Saeed (2025), drawing on the theoretical framework of Barro (1991). They include: gross savings as a share of GDP (a proxy for gross capital formation and the investment channel); trade openness (the sum of exports and imports as a share of GDP); the inflation rate (capturing macroeconomic stability); net foreign direct investment inflows as a share of GDP; population growth; a binary conflict indicator equal to one in country-years where battle-related deaths exceed 1,000, following the UCDP operational threshold for war; and total natural resource rents (the sum of oil, natural gas, coal, mineral and forest rents as a percentage of GDP), allowing one to isolate the effect of military expenditure from resource-driven growth dynamics. A log-transformed, one-period lagged GDP per capita (constant 2015 USD) is added as a conditional convergence term following Barro (1991), who predicts that countries further from their steady-state income level tend to grow faster; the associated coefficient is negative and highly significant, consistent with conditional convergence. All variables are collected from the World Bank World Development Indicators (WDI).

4.2 Econometric Specification

The baseline empirical specification, used by Bishnoi et al. (2025) and Dimitriou et al. (2024), is a Two-Way Fixed Effects (TWFE) panel model of the following form:

$$\text{GDPgrowth}_{it} = \alpha_i + \lambda_t + \beta \times \text{miles}_{i,t-1} + \gamma' X_{it} + \varepsilon_{it} \quad (1)$$

where GDPgrowth_{it} denotes the GDP growth rate (constant 2015 USD) of country i in year t ; α_i are country fixed effects, absorbing all time-invariant heterogeneity across countries; λ_t are year fixed effects, capturing common global shocks; $\text{miles}_{i,t-1}$ is the military burden lagged by one period; β is the coefficient of interest, capturing the within-country effect of a one-percentage-point change in the military burden on growth the following year; and X_{it} is a vector of time-varying control variables described in Section 4.1.

The use of a one-period lag on the military burden is motivated by reverse causality concerns: contemporaneous GDP growth may mechanically affect the military burden, and a country experiencing a growth shock may simultaneously revise its defense allocations. Lagging the key independent variable by one year breaks this simultaneity and provides a cleaner identification of the direction of causation from military expenditure to growth. As a robustness check, results using the contemporaneous value of the military burden were also computed: the sign and significance of the main coefficient are unchanged. Standard errors are clustered at the country level, allowing for arbitrary serial correlation and heteroskedasticity within countries over time (Arellano, 1987).

5. Empirical Results

5.1 An Inconclusive but Revealing Aggregate Result

Taken as a whole, the Asia-Pacific region does not yield a single clear verdict on the effect of military spending on growth. Using the TWFE regression on Asia-Pacific countries between 1993 and 2024, the estimated effect of the military burden on GDP growth is negative but non-significant, with a coefficient of -0.122 and an explanatory power (within R^2) of 17.9%.⁹ This means that, on average, a one percentage point increase in the military burden is associated with a 0.122 percentage point reduction in GDP growth. To put this in perspective: for a country growing at 5 per cent per year, this would imply a reduction to approximately 4.88 per cent if the

⁹The within R^2 of 0.179 means that 17.9% of the temporal variation in each country is explained by the model.

military burden rose by one percentage point. However, this result is not statistically significant enough to be conclusive on its own, although it leaves insightful clues about the crowding-out effect of the military burden (Column 1 of Table 1).

This ambiguity is not a failure of the analysis. It reflects the profound heterogeneity of the Asia-Pacific region, which encompasses economies as different as Singapore and Tajikistan, Australia and Myanmar, Japan and Afghanistan. These countries differ in their industrial structures, income levels, institutional capacities, security environments, and degrees of integration into global trade and financial markets.

The inconclusive aggregate result appears consistent with the literature: some countries may benefit from defense-related industrial activity, while others face severe crowding-out with no offsetting gains. When these opposing effects are averaged together across the full sample, they partially cancel each other, producing the non-significant aggregate result observed. The following sub-sections disaggregate the results by arms export status and conflict exposure to explore the military burden-growth nexus more precisely.

5.2 Arms Export Status as a Moderating Factor

The negative effect of military spending on growth is concentrated among countries that do not export arms, while for major exporters the defense sector generates enough industrial activity to absorb its own fiscal cost.

A clearer picture emerges when countries are distinguished by their arms export intensity. Each year, countries are classified as top 10% Asia-Pacific arms exporters if their TIV/GDP ratio¹⁰ ranks in the top quantile, using a three-year rolling average to smooth delivery spikes.¹¹ An interaction term between the military burden and the top-exporter category then allows for exploration of the differential effect between major exporters and all other countries.

The results are striking. For countries that are not major arms exporters, the majority of Asia-Pacific states over the period, a one percentage point increase in the military burden is associated with a statistically significant decrease of 0.293 percentage points in GDP growth (Column 2 of Table 1). This is a meaningful economic penalty. By contrast, for countries that are top arms exporters, the same increase in military burden carries an additional 0.930 percentage points effect compared to the non-exporter effect.¹² This produces a total spending effect of

¹⁰SIPRI Trend Indicator Values (TIV) constitute a standardised measure of the volume of weapons transferred.

¹¹The fact that countries can enter and exit the top 10% prevents perfect collinearity with country fixed effects.

¹²Countries classified as top exporters for at least 8 years over the 1993-2024 period are: China, Russia, Rep. of Korea, Singapore, Türkiye, Kyrgyz Republic, Uzbekistan, Kazakhstan and Georgia.

+0.637 percentage points for major arms-exporting countries.

This indicates that the crowding-in mechanism operates as the theoretical literature predicts (Callado-Muñoz et al., 2022; Dimitriou et al., 2025; Rahman and Siddiqui, 2019). The defense sector in these economies generates sufficient industrial spillovers, economies of scale in weapons production, and export revenues to totally offset and even overcome the fiscal cost of higher military spending. Thus, not only does arms export neutralise the cost of rearmament, but it turns it into an economic asset for greater militarisation.

5.3 Conflict Exposure as a Moderating Factor

In countries already at war, military spending does not harm growth; rather, it helps stabilise the security environment and sustains domestic demand when private economic activity has collapsed.

A separate and important pattern emerges for conflict-affected countries.¹³ Being in active conflict is itself associated with a severe reduction in GDP growth, reflecting the broad disruption that war inflicts on supply chains, investment flows, trade, and economic confidence (Aziz and Khalid, 2019). In concrete terms, armed conflicts deeply disturb the normal functioning of an economy: businesses close, workers flee, infrastructure is destroyed, and foreign investors withdraw. Poverty rates in Myanmar, for instance, nearly doubled from 24.8 per cent in 2017 to 49.7 per cent in 2023 due to the ongoing civil war (UNDP, 2024).

Within this already difficult context, the effect of military spending itself changes. During peaceful years, a one percentage point increase in the military burden is associated with a non-significant decrease of 0.169 percentage points in GDP growth. During conflict years, however, the picture reverses. The additional spending effect during conflict is positive and statistically significant at the 5 per cent level, estimated at 0.495 percentage points (relative to the peaceful years effect), producing a total spending effect during conflict of 0.326 percentage points, significant at the 10 per cent level (Column 3 of Table 1).

Two complementary mechanisms explain this reversal. First, higher military spending in conflict settings can help stabilise the security environment by funding armed forces, securing trade routes, and protecting civilian infrastructure, thereby restoring a degree of confidence among businesses and investors. Second, defense expenditure generates immediate domestic demand through personnel wages and procurement from local suppliers, which can partly offset the contraction of the

¹³Countries included in the conflict sub-sample are: Afghanistan, India, Sri Lanka, Pakistan, Türkiye, Russia, Myanmar, Nepal, Philippines, Azerbaijan, Tajikistan and Georgia.

private sector. When private economic activity collapses due to insecurity, government military spending becomes one of the few remaining sources of domestic demand.

However, this stabilising effect operates only conditional on conflict already occurring: it does not imply that higher military spending prevents conflicts from breaking out in the first place. Indeed, even if a deterrence effect can be observed, sharp increases in military spending yield only modest preventive gains (Benmelech and Monteiro, 2025). Thus, the core message remains: for the vast majority of Asia-Pacific countries not currently at war, the costs of higher military spending are real and are not offset by compensating mechanisms. Higher military spending is only growth-positive when a country is already paying the far greater cost of an active conflict.

Table 1: Panel Regression Results: Military Burden and GDP Growth (TWFE)

	(1) Baseline	(2) Arms Exporter	(3) Conflict
<i>Military burden_{t-1}</i>	-0.122		
<i>Military burden_{t-1} (non-exporters)</i>		-0.293**	
<i>Military burden_{t-1} × Top-exporter (interaction)</i>		+0.930**	
<i>[Total effect for top exporters]</i>		[+0.637]*	
<i>Military burden_{t-1} (peaceful years)</i>			-0.169
<i>Military burden_{t-1} × Conflict (interaction)</i>			+0.495**
<i>[Total effect during conflict]</i>			[+0.326]*
<i>lnGDP per capita (lagged)</i>	-4.58***	-4.67**	-4.54***
<i>Natural rents</i>	+0.16***	+0.16**	+0.16***
Controls	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Within R^2	0.179	0.187	0.180
Period	1993-2024	1993-2024	1993-2024

Notes: Dependent variable: annual GDP growth rate (constant 2015 USD). Military burden is measured as military expenditure as a percentage of GDP, lagged by one year. Top arms exporters are countries ranked in the top 10% of Asia-Pacific member states by TIV/GDP (three-year rolling average). Conflict is defined as country-years with more than 1,000 battle-related deaths per year (UCDP threshold). Standard errors are clustered at the country level (Arellano, 1987). Controls include: gross savings/GDP, trade openness, inflation, net FDI/GDP, population growth, natural resource rents/GDP, and log GDP per capita (lag). Year 2020 is excluded. Brackets denote linear combinations of coefficients.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

6. Discussion

The empirical results confirm the fundamental insight emerging from the meta-analytical literature: the relationship between military spending and economic growth is deeply context-dependent, and uniform policy conclusions are therefore misleading. Three core findings emerge from the analysis, each with direct implications for how Asia-Pacific governments should calibrate their defense and fiscal strategies.

The non-significant aggregate result for the Asia-Pacific region as a whole is not a statistical failure but an accurate reflection of structural heterogeneity. The region encompasses economies at very different stages of industrial development, exposed to different security environments, and with fundamentally different capacities to translate defense spending into productive economic activity. Averaging across these dimensions necessarily produces a muted, inconclusive estimate. The relevant analytical questions are therefore at a disaggregated level.

6.1 The Asymmetric Role of Arms Export Capacity

Arms export status is the single most important moderating factor identified in this analysis. The gap between the effect for non-exporters (-0.293 pp) and for major exporters ($+0.637$ pp) represents a difference of nearly one full percentage point in growth per unit of military burden, an economically substantial magnitude. This finding is consistent with the crowding-in mechanism described by Callado-Muñoz et al. (2022) and Moretti et al. (2025): when the defense sector generates industrial spillovers and export revenues, it can absorb and even reverse the fiscal cost of higher military spending. However, only a small number of Asia-Pacific countries meet the conditions for this mechanism to operate. The majority of Asia-Pacific states are net importers of arms, lack significant military-industrial complexes, and face the full crowding-out effect without compensating returns.

This asymmetry calls for differentiated strategies. Empirical evidence reveals a sharp divide between the handful of substantial Asia-Pacific arms exporters and the rest of the region. These economies appear able to maintain economic growth alongside higher military burdens, as the defense sector generates sufficient industrial spillovers and export revenues to absorb the fiscal cost. For arms net-importing economies, especially those without an important military-industrial complex, the priority should be to minimise the cost of defense spending. For economies with a clear defense industrial capacity, the military burden can be partially increased, but only if accompanied by investment in the civilian innovation ecosystems as universities, technology transfer mechanisms, and start-up ecosystems capable of absorbing and commercialising defense-generated knowledge. These conditions are

met by only a small number of Asia-Pacific member states, and claims of R&D spillover benefits should not be used to justify a broad increase in military expenditure across the region.

Two complementary mechanisms can attenuate the cost asymmetry between importers and exporters. First, cross-border defense-industrial partnerships create financial and industrial interdependence that makes military confrontation between partners economically irrational. The logic is not new: the European Coal and Steel Community, established in 1951, made Franco-German confrontation structurally impossible by pooling the raw materials from which weapons are made. Agreements such as the Malaysian–Korean one, whereby Malaysia serves as a regional maintenance hub for the FA-50 aircraft, should be reproduced by other Asia-Pacific member states. Central Asian and Pacific Island states, which hold significant deposits of rare earth elements and strategic metals, are natural candidates for formalised supply and technology-transfer agreements with exporting states. Second, public-private partnerships in defense can draw private capital rather than displacing it and generate civilian innovation spillovers. By making private firms and universities active participants in defense investments, these partnerships allow private capital to enter the defense sector voluntarily and co-finance research and development, with the resulting technologies diffusing into civilian industry through commercialisation rights. The Republic of Korea illustrates this model most concretely: Samsung’s entry into the defense sector has turned military spending into industrial upgrading. Asia-Pacific states should promote joint research agreements between defense ministries, universities, and private conglomerates. The Asian Development Bank could extend this model further by establishing conditional financing solutions for dual-use technology partnerships.

6.2 Conflict, Security, and the Primacy of Peace

The conflict-state conditionality of military spending’s effect underscores the primacy of peace as an economic asset. The positive short-run effect of military spending during active conflict, operating through demand stabilisation and security restoration, does not overturn this conclusion. Rather, it illustrates the logic of lesser evils: military spending becomes growth-positive only because the counterfactual of no military response to an active conflict carries an even larger economic cost. The evidence of Benmelech and Monteiro (2025) further tempers the deterrence argument: preventive military spending yields only modest reductions in the probability of conflict, calling into question the economic efficiency of arms build-ups as conflict-prevention tools.

For countries already engaged in an active conflict, higher defense spending can generate a positive effect on GDP growth through two channels. On the demand side,

military expenditure creates direct stimulus through personnel wages and procurement of goods, sustaining domestic economic activity even when private-sector demand has collapsed. On the security side, a stronger military posture can help stabilise the conflict environment, protect trade routes and civilian infrastructure, and gradually restore the conditions needed for private investment to return. Conflict-affected Asia-Pacific governments should therefore increase their military burden if necessary, while simultaneously pursuing diplomatic solutions. The economic recovery ultimately depends on the restoration of peace, not on high defense spending, which can only partially offset the negative economic effects of conflicts. Preventing conflicts is therefore the single most effective growth-enhancing policy available.

6.3 The Collective Action Problem and the Responsibilities of Major Powers

The post-2014 trajectory of the Asia-Pacific region, characterised by a simultaneously rising military burden and a deceleration of regional GDP growth tracked by the geopolitical risk index (Caldara and Iacoviello, 2022), is consistent with these findings. The action-reaction dynamic documented by the United Nations Secretary-General (2024) generates a collectively welfare-reducing arms spiral, in which the aggregate fiscal cost is borne disproportionately by the smaller and lower-income Asia-Pacific states who lack the industrial capacity to offset it.

This spiral is largely driven by the choices of major regional powers as China, Russia, India, Iran, Pakistan, and Türkiye, whose military build-ups set the pace of regional rearmament and cascade through the security calculations of smaller neighbours. These countries also happen to be, with the exception of India and Pakistan, major arms exporters, meaning they benefit economically from rising regional tensions: higher domestic military spending expands their defense industries and international demand for arms exports. This advantage comes, however, at a significant collective cost. As major powers drive up the regional military burden, smaller and lower-income Asia-Pacific states are compelled to follow, diverting fiscal resources away from education, health, infrastructure, and the investments needed to meet the Sustainable Development Goals. Evidence shows that a higher military burden might deter and reduce conflicts within the region, but this impact is relatively weak and also leads to a higher cost of war, greater casualties, and greater economic destruction. Preventing the rise in geopolitical tensions and militarisation is therefore not only a security issue but also an economic one, and a global de-escalation of the military burden constitutes the most effective collective policy response available to the region.

7. Conclusion

This paper has examined the economic implications of rising geopolitical rivalry for the Asia-Pacific region through a combined review of the academic literature, a historical documentation of peace dividends, and panel econometric analysis of Asia-Pacific member states over 1993-2024.

The evidence points to three main conclusions. First, the aggregate effect of military spending on economic growth across the Asia-Pacific region is negative but statistically non-significant, reflecting the deep heterogeneity of a region that encompasses both advanced industrial economies and low-income developing countries. Second, this heterogeneity is structured and predictable: the negative effect is concentrated among non-arms-exporting countries, which constitute the majority of Asia-Pacific states and bear the full crowding-out cost of higher military burdens without offsetting industrial or export benefits. Major arms exporters, by contrast, experience net positive effects, consistent with the crowding-in mechanism documented in the literature. Third, during active conflict, higher military spending can sustain economic activity through demand and security channels, but this finding must be read alongside the much larger negative effect of conflict itself, reinforcing the primacy of conflict prevention as an economic priority.

These findings carry direct implications for the trajectory that the Asia-Pacific region is currently on. The post-2014 intensification of geopolitical rivalries has generated an action-reaction dynamic in military spending that is collectively welfare-reducing, particularly for the smaller and lower-income countries that lack the industrial capacity to offset its fiscal cost. The opportunity cost is not abstract: it is measured in the financing gap for the Sustainable Development Goals, in social expenditure foregone, and in investment not made.

The path forward requires differentiated strategies, between importers and exporters, between peaceful and conflict-affected countries, and between those with and without the institutional capacity to generate civilian spillovers from defense investment. Above all, it requires a recognition that the peace dividend of the past three decades was not an accident but an achievement, and that its erosion today carries real and quantifiable economic costs for the region's development prospects.

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